

Experiment 15 – Custom Keywords and Reusable Test Automation

1. Module Name and Experiment Title

Module Name: Module 4 – Robot Framework

Experiment Number: Experiment 15

Experiment Title: Custom Keywords and Reusable Test Automation

Programming Language: Python (Robot Framework)

Website: The Internet – Login Page

Website URL: <https://the-internet.herokuapp.com/login>

2. Problem Statement

To develop a Robot Framework test automation script using custom keywords to make test cases more readable, reusable, and maintainable. The experiment automates the login process of a web application by creating reusable keywords for opening the browser, navigating to the login page, entering credentials, and verifying successful login.

3. Objective

- To understand how to create and use custom keywords in Robot Framework.
 - To improve test script readability by dividing test steps into reusable keywords.
 - To automate the login process of a web application.
 - To use SeleniumLibrary for browser automation.
 - To verify successful login using assertions.
 - To generate execution reports and logs using Robot Framework.
-

4. Tools, Software and Concepts Used

Tools and Software:

- Python
- Robot Framework
- SeleniumLibrary
- Google Chrome
- Visual Studio Code
- The Internet web application

Concepts Used:

- Custom keywords
- Keyword-driven test automation
- Selenium browser automation
- Test setup and teardown
- Element locators
- Assertions and validation
- Test execution reports and logs

5. Source Code / Implementation Steps

A. Project Structure

Experiment 15 - Custom Keywords and Reusable Test Automation

```

├── source_code
│   ├── test_custom_keywords.robot
│   └── requirements.txt
├── screenshots
│   ├── 01_successful_login.png
│   └── 02_login_after_logout.png
└── reports
    ├── log.html
    ├── output.xml
    └── report.html
  
```

B. requirements.txt

```

robotframework
robotframework-seleniumlibrary
  
```

C. Robot Framework Test Script

File Name: test_custom_keywords.robot

```

*** Settings ***
Library      SeleniumLibrary

Suite Setup      Open Test Browser
Suite Teardown   Close Browser

*** Variables ***
${URL}           https://the-internet.herokuapp.com/login
${USERNAME}      tomsmith
${PASSWORD}      SuperSecretPassword!
${BROWSER}       chrome

*** Test Cases ***
Verify Successful Login Using Custom Keywords
    [Documentation]    Verify successful login using reusable custom
keywords.
    Open Login Page
  
```

```

    Enter Login Credentials
    Submit Login Form
    Verify Secure Area
    Capture Page Screenshot    ../screenshots/01_successful_login.png
    Log    Login successful and verified.

*** Keywords ***
Open Test Browser
    Open Browser    about:blank    ${BROWSER}
    Maximize Browser Window

Open Login Page
    Go To    ${URL}
    Wait Until Element Is Visible    id=username    timeout=10s

Enter Login Credentials
    Input Text    id=username    ${USERNAME}
    Input Text    id=password    ${PASSWORD}

Submit Login Form
    Click Button    css=button[type="submit"]

Verify Secure Area
    Wait Until Page Contains    You logged into a secure area!
    timeout=10s
    Page Should Contain    You logged into a secure area!

```

D. Implementation Steps

1. Install Python and Robot Framework.
2. Install SeleniumLibrary using the `requirements.txt` file.
3. Create the `test_custom_keywords.robot` file inside the `source_code` folder.
4. Import SeleniumLibrary in the Settings section.
5. Define the URL, username, password, and browser as variables.
6. Create custom keywords for opening the browser, navigating to the login page, entering credentials, submitting the form, and verifying the secure area.
7. Execute the test case using the Robot Framework command.
8. Check the generated output, log, and report files.

E. Installation and Execution Commands

Install the required packages:

```
pip install -r source_code/requirements.txt
```

Run the test script from the experiment folder:

```
robot --outputdir reports source_code/test_custom_keywords.robot
```

6. Output with Clearly Labelled Screenshots

Screenshot 1: Successful Login

● You logged into a secure area!

Secure Area

Welcome to the Secure Area. When you are done click logout below.

Logout

Powered by [Elemental Selenium](#)

Post me on GitHub

Screenshot 2: Login Page After Logout

● You logged out of the secure area!

Login Page

This is where you can log into the secure area. Enter *tomsmith* for the username and *SuperSecretPassword!* for the password. If the information is wrong you should see error messages.

Username

Password

➔ Login

Powered by [Elemental Selenium](#)

Post me on GitHub

Screenshot 3: Robot Framework Execution Log

Summarize Point SixArpan Pal_12023002028083/iPractice PageFramesTest Custom Keywords LogAsk Gemini

C:/Users/rajar/Desktop/Selenium/Codes%20for%20lab%20work/Module%20-%204/Experiment%2015%20-%20Custom%20Keywords%20and%20Reusable%20Test%20Automation/reports...

Test Custom Keywords Log

Generated20260927 23:02:41 UTC+05:3045 seconds ago

REPORT

Statistics

Total Statistics	Total	Pass	Fail	Skip	Elapsed	Pass / Fail / Skip
All Tests	1	1	0	0	00:00:06	

Statistics by Tag	Total	Pass	Fail	Skip	Elapsed	Pass / Fail / Skip
No Tags						

Statistics by Suite	Total	Pass	Fail	Skip	Elapsed	Pass / Fail / Skip
Test Custom Keywords	1	1	0	0	00:00:10	

Execution Log

SUITE Test Custom Keywords00:00:10.396

Full Name:Test Custom Keywords

Source:C:/Users/rajan/Desktop/Selenium/Codes for lab work/Module - 4/Experiment 15 - Custom Keywords and Reusable Test Automation/source_code/test_custom_keywords.robot

Start / End / Elapsed:20260927 23:02:31.526 / 20260927 23:02:41.922 / 00:00:10.396

Status:1 test total, 1 passed, 0 failed, 0 skipped

SETUP Open Test Browser00:00:02.270

TEARDOWN SeleniumLibrary.Close Browser00:00:02.179

TEST Verify Successful Login Using Custom Keywords00:00:05.550

28°CPartly cloudy23:0327-09-2026

Screenshot 4: Robot Framework Execution Report

Summarize Point SixArpan Pal_12023002028083/iPractice PageFramesTest Custom Keyword:Test Custom Keyword:Ask Gemini

C:/Users/rajar/Desktop/Selenium/Codes%20for%20lab%20work/Module%20-%204/Experiment%2015%20-%20Custom%20Keywords%20and%20Reusable%20Test%20Automation/reports...

Test Custom Keywords Report

Generated20260927 23:02:41 UTC+05:301 minute 1 second ago

LOG

Summary

Status:All tests passed

Start Time:20260927 23:02:31.526

End Time:20260927 23:02:41.922

Elapsed Time:00:00:10.396

Log File:log.html

Statistics

Total Statistics	Total	Pass	Fail	Skip	Elapsed	Pass / Fail / Skip
All Tests	1	1	0	0	00:00:06	

Statistics by Tag	Total	Pass	Fail	Skip	Elapsed	Pass / Fail / Skip
No Tags						

Statistics by Suite	Total	Pass	Fail	Skip	Elapsed	Pass / Fail / Skip
Test Custom Keywords	1	1	0	0	00:00:10	

Details

AllTagsSuitesSearch

Suite:

Test:

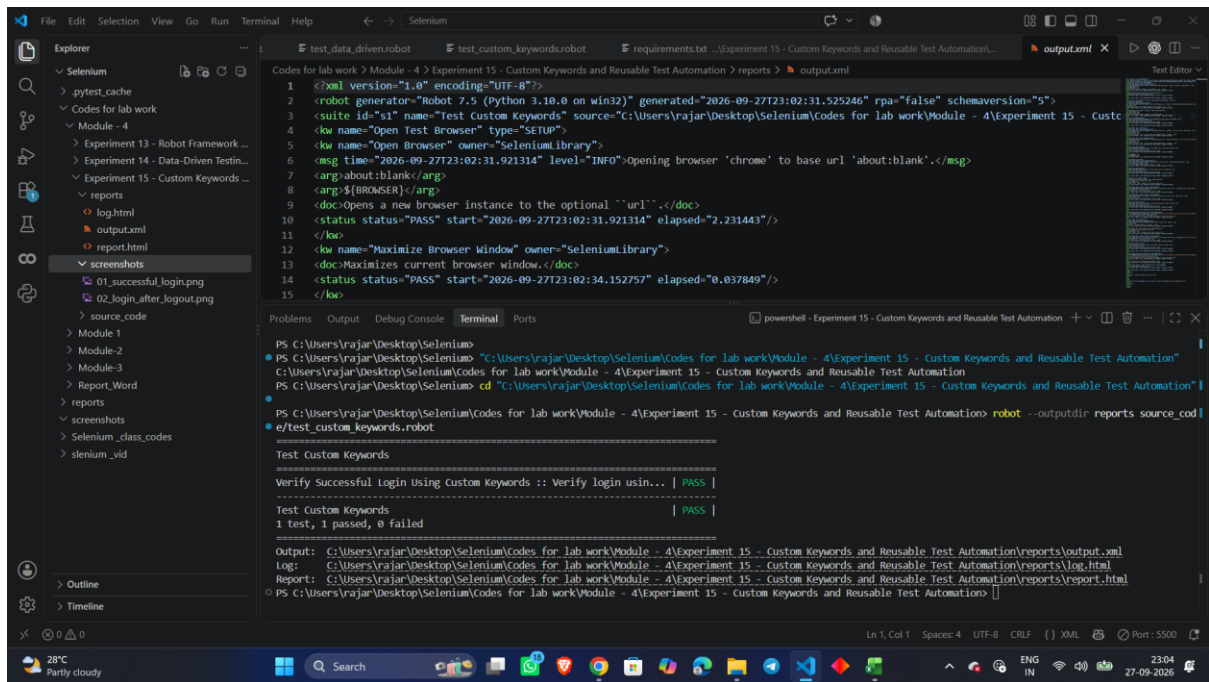
Include:

Exclude:

SearchClearHelp

28°CPartly cloudy23:0327-09-2026

Screenshot 5: Terminal Execution Output



Execution Result

Parameter	Result
Test Case	Verify Successful Login Using Custom Keywords
Total Tests	1
Passed	1
Failed	0
Skipped	0
Execution Status	PASS
Execution Time	Approximately 10 seconds

7. Result, Observation and Conclusion

The experiment was successfully completed using Robot Framework and SeleniumLibrary. Custom keywords were created to divide the login automation process into reusable steps. The test successfully opened the login page, entered valid credentials, submitted the login form, and verified that the Secure Area was displayed.

The execution report confirmed that **1 test passed, with 0 failures and 0 skipped tests**.

This experiment demonstrated how custom keywords can improve the readability, reusability, and maintainability of Robot Framework test scripts.